



CL-2001 Data Structures Lab # 9

Objectives:

1. AVL Tree

Note: Carefully read the following instructions (*Each instruction contains a weightage*)

1. There must be a block of comments at the start of every question's code by students; the block should contain brief description about functionality of code.
2. Comment on every function about its functionality.
3. Use understandable names of variables.
4. Proper indentation of code is essential.
5. Write a C++ statement(s) for each of the following tasks one after the other, in the same order.
6. Make a Microsoft Word file and paste all of your C++ code with all possible screenshots of every **task output in MS word and submit .cpp file with word file.**
7. Make separate .cpp files for all tasks and use this format **22F-1234_Task1.cpp.**
8. First think about statement problems and then write/draw your logic on copy.
9. After copy pencil work, code the problem statement on MS Studio C++ compiler.
10. At the end when you done your tasks, attached C++ created files in MS word file and make your submission on Google classroom. (Make sure your submission is completed).
11. Please submit your file in this format **22F-1234_L1.**
12. Do not submit your assignment **after the deadline.**
- 13. Do not copy code from any source otherwise you will be penalized with negative marks.**



Problem 1:

Provide a C++ implementation of AVL tree which must include

- Recursive Height
- Finding Balancing Factor
- RR_Rotation(node* Parent)
- LL_Rotation(node* Parent)
- RL_Rotation(node* Parent)
- LR_Rotation(node* Parent)
- Apply on BST deletion
- Apply on BST insertion
- Display Nodes
- Test Your Code

Problem 2:

Write a recursive function to check if the provided tree is a valid AVL tree or not.

Problem 3:

Write a C++ program to convert a provide BST into a AVL tree.

Problem 4:

Write a C++ function which is passed two AVL trees T1 and T2, where the largest key in T1 is less than the smallest key in T2, Join(T1, T2) returns an AVL tree containing the union of the elements in T1 and T2.